It is because of the two $E_1, E_2$ permuted internally will give rise to the same permutations. Since they are same, the required number of permutations is $\frac{3!}{2!} = 3$.

**Theorem 4.4:** The number of permutations of $n$ objects, where $p$ objects are of the same kind and rest are all different is $\frac{n!}{p!}$.

Generally, the number of permutations of n objects, where $p_1$ objects are of one kind, $p_2$ objects are of second kind, $\cdots$ $p_k$, are of $k^{th}$ kind and the rest of it are of different kind is $\frac{n!}{p_1! \times p_2! \times \cdots \times p_k!}$.

**Example 4.36** Find the number of ways of arranging the letters of the word BANANA.

**Solution:**
This word has 6 letters in which there are 3 A'S, 2 N's and one B. The number of ways of arrangements is $\frac{6!}{3! \times 2!} = 60$.

**Example 4.37** Find the number of ways of arranging the letters of the word RAMANUJAN so that the relative positions of vowels and consonants are not changed.

**Solution:**
In the word RAMANUJAN there are 4 vowels (A,A,U,A)in that 3 A's, 1 U and 5 consonants (R,M,N,J,N) in that two N's and rest are distinct. The 4 vowels (A,A,A,U) can be arranged themselves in $\frac{4!}{3!} = 4$ ways. The 5 consonants (R,M,N,J,N) can be arranged themselves in $\frac{5!}{2!} = 60$ ways. Therefore the number of required arrangements are $\frac{4!}{3!} \times \frac{5!}{2!} = 4 \times 60 = 240$.

**Example 4.38** Three twins pose for a photograph standing in a line. How many arrangements are there (i). when there are no restrictions. (ii). when each person is standing next to his or her twin?

**Solution:**

(i) The six persons without any restriction may be arranged in ${}^6P_6 = 6! = 720$ ways.
(ii) Let us consider three twins as $T_1, T_2, T_3$. Each twin is considered as a single unit and these three can be permuted in 3! ways. Again each twin can be permuted between themselves in 2! ways. Hence, the total number of arrangements is $3! \times 2! \times 2! \times 2! = 48$ ways.

**Example 4.39** How many numbers can be formed using the digits 1,2,3,4,2,1 such that, even digits occupies even place?

**Solution:**
There are 6 places in that there are 3 even places we have 2,4,2 as even numbers. The number of ways of permuting 2,4,2 in the 3 even places in $\frac{3!}{2!} = 3$ ways. The remaining numbers 1,3,1 can be permuted in the remaining 3 places in $\frac{3!}{2!} = 3$ ways. Hence, the required number of numbers is $3 \times 3 = 9$.